LARSI-TPE-XGB: Short-Term Load Forecasting by Load-Adaptive Relative Strength Index and Fusion of Tree-Structured Parzen Estimator and XGBoost

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***Abstract*—Power load forecasting is essential for optimizing power generation and distribution efficiency. This paper proposes a novel method for daily average load forecasting, referred to as LARSI-TPE-XGB, which integrates the Load-Adaptive Relative Strength Index (LARSI) with the Tree-structured Parzen Estimator (TPE) and eXtreme Gradient Boosting (XGBoost). Our method significantly improves the accuracy and generalization ability of short-term load forecasting (STLF) by addressing limitations in feature extraction and hyperparameter optimization. The proposed LARSI enhances the forecasting model by adapting an improved Relative Strength Index (RSI) for power load prediction, while TPE optimizes the model's hyperparameters to dynamically adjust to time-series updates, thus mitigating the issue of XGBoost's sensitivity to hyperparameters in high-dimensional scenarios. Experimental results on real-world power load datasets demonstrate that LARSI-TPE-XGB reduces errors by 18.58% and 30.73% in root mean squared error (RMSE) across two different datasets compared to models without LARSI-TPE-XGB and outperforms state-of-the-art models, as confirmed by the Diebold-Mariano (DM) test. Our method consistently improves performance across various datasets, while we further investigate the influence of LARSI and other factors, such as weather conditions, on forecasting accuracy.**

***Index Terms*—Hyperparameter optimization, load-adaptive relative strength index, machine learning, power load forecasting, short-term load forecasting.**

I. Introduction

FORECASTING power load plays a crucial role in enabling power companies to make strategic decisions regarding infrastructure expansion, electricity procurement, and demand management. Reliable forecasts are essential for maintaining stable electricity generation and reducing the risk of power outages [1]. In addition, they help transmission system operators (TSOs) optimize system planning and improve transmission efficiency [2]. The US grid operator, PJM, states that precise short-term forecasts enable power companies to economically dispatch power plants, balance supply and demand, and substantially improve the performance of dispatch systems, leading to cost savings [3]. The interconnectedness of electrical grids across European countries has made cross-border dispatch possible, and accurate forecasting ensures the efficiency and effectiveness of such cross-border dispatch [4]. Therefore, precise load forecasting not only helps prevent power shortages but also optimizes resource allocation, leading to cost reductions.

Load forecasting can be divided into different time frames, including long, medium, and short-term predictions. Long- and medium-term forecasts typically extend for several weeks to several years and are essential for strategic planning and accounting for seasonal variations in demand. Conversely, short-term load forecasting focuses on forecasting loads from just minutes to one week ahead. Short-term load forecasting (STLF) plays an important role in the daily management of the power grid [5].

Many studies on STLF have focused on day-ahead hourly load forecasting [6]. This approach helps in understanding demand at different times of day. However, it is susceptible to short-term fluctuations and the accumulation of errors due to noise from uncontrollable factors, resulting in significant errors [7]. Compared to hourly forecasting, daily average forecasting is more suitable for larger-scale planning and decision-making, as it offers smoother daily averages [8]. Average power load forecasting on a daily cycle can reduce scheduling and computational complexity [9]. For utility companies, accurate daily average forecasting streamlines scheduling and minimizes adjustments. Electricity traders can focus on macro strategies while mitigating short-term volatility. Thermal power plants can better estimate daily fuel needs, thereby reducing fluctuations and simplifying procurement. Notable contributions in this field include research on daily average load forecasting in Guiyang, China [10]. Based on these works, this paper proposes a novel method for daily average load forecasting, with the aim of improving the accuracy and adaptivity of existing forecasting methods.

Different types of nonstationary processes can exist in electricity loads [11]. However, most current forecasting methods rely on the assumptions that time series data exhibits stationarity or that this data can be made stationary by eliminating trends and seasonal fluctuations [12]. Therefore, feature extraction is an important feature of load forecasting, as it allows forecasting models to identify load patterns, significantly improving accuracy. Some studies incorporate electricity prices, numerical weather predictions (NWPs), and calendar features into load forecasting [13], while others derive features from historical information [14].

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TABLE I
OUR CORE CONTRIBUTIONS COMPARED TO EXISTING WORKS

Contribution	Our proposed work	Existing works
Enhancing Load Forecasting Features (LARSI)	We present the novel Load Adaptive Relative Strength Index (LARSI), an innovative adaptation of the financial RSI indicator, to significantly enhance load forecasting. By integrating the proven short-term financial predictive power of RSI into load forecasting, we pioneer a groundbreaking approach that promises improved accuracy.	While most studies rely on traditional inputs like lag values, weather data, and calendar-based features, only a few have pioneered innovative and effective features that significantly enhance forecasting accuracy.
Advanced Feature Selection and Model Adaptation (TPE-XGB)	We introduce TPE-XGB, a powerful integration of XGBoost [16], [17] and the Tree-structured Parzen Estimator (TPE). XGBoost efficiently manages high-dimensional features and excels in forecasting, especially when combined with LARSI. TPE automatically tunes hyperparameters, resolving common sensitivity issues in machine learning models and ensuring continuous optimization as new data arrives. This combination enhances both accuracy and adaptability, making TPE-XGB an outstanding solution for load forecasting.	Many studies have utilized deep learning for automatic feature learning and forecasting [18], [19], which often requires large volumes of data. This approach can be limiting in daily average load forecasting. Additionally, existing research often overlooks the need for model adaptation in real-world applications, where forecasting models should be updated over time. The issues of maintaining model relevance over time and ensuring adaptability are frequently not addressed.

Recent research has focused primarily on enhancing prediction model architectures, with relatively little emphasis on feature development. In our initial study, presented as a conference paper [15], we introduced the novel application of the standard Relative Strength Index (RSI) from stock market analysis to STLF. This study substantially extends our earlier research by proposing the Load-Adaptive RSI (LARSI), a more refined and versatile version of the RSI specifically tailored for load forecasting. Unlike the standard RSI, LARSI integrates exponential moving average (EMA) and Savitzky-Golay (SG) smoothing techniques, making it better suited for capturing distinct patterns in load data. Additionally, we expanded the analysis by performing rigorous comparisons with existing forecasting models using Diebold-Mariano (DM) tests [20], investigating the influence of weather data on load prediction, and validating the effectiveness of LARSI across three distinct datasets. Furthermore, we conducted a thorough feature importance analysis to better understand the contribution of individual features to forecasting accuracy and examined the model's convergence behavior to ensure its stability and robustness—enhancements not addressed in our preliminary work.

Moreover, although deep learning (DL) has shown recent promising performance in time series prediction [21], it requires a large amount of data to automatically extract features from a time series, which is challenging to implement in daily load forecasting. The volume of time series data is limited by the period from the start of data collection to the present. In contrast, other prediction tasks, such as image classification, can quickly gather larger volumes of data by allocating more resources in a short time. Most load datasets only contain data spanning a few years, as evidenced by [7] and [22].

To effectively capture the characteristics of load time series, this paper introduces LARSI and proposes TPE-XGB, a novel integration of TPE [23] and XGBoost [24]. XGBoost, a powerful model for contemporary prediction tasks, possesses automatic feature selection capabilities, enabling it to work with a significant number of predefined features. However, when used with high-dimensional features, XGBoost and other machine learning (ML) models exhibit sensitivity to hyperparameter settings [25], critically affecting prediction performance. Furthermore, as time series data continuously evolve, periodic updates may be necessary to mitigate changes in the optimal hyperparameters [26]. Accordingly, we propose the use of TPE for hyperparameter optimization to dynamically adapt to changes in time series data, thereby enhancing the performance of STLF. This approach addresses the limitations of manual tuning, which can be cumbersome and time-consuming, an aspect often overlooked in many research studies on load forecasting despite its significant impact on model performance.

As our method combines our novel LARSI index with our novel TPE-XGB forecasting method, we refer to our proposed method as LARSI-TPE-XGB. Our core contributions compared to existing works are shown in Table I.

In addition, the main distinguishing features of the proposed scheme, compared to similar methods, are shown in Table II. Compared with the other two primary approaches, LARSI-TPE-XGB combines the advantages of traditional machine learning and dynamic optimization. Given that time series data are often limited—making it difficult to collect significantly more data in a short period—and that overly distant historical data may be discarded due to quality or relevance concerns (thus further restricting usable data), LARSI-TPE-XGB requires relatively low data while enhancing feature engineering through LARSI. It also employs TPE to automatically tune XGBoost hyperparameters, effectively adapting to load fluctuations. Furthermore, by incorporating time-based dynamic updates, this approach exhibits greater flexibility and accuracy in real-world applications.

The structure of the remainder of this paper is as follows: Section II details the methodologies related to the proposed approach. Section III provides a detailed explanation of the proposed LARSI-TPE-XGB. Section IV details the experimental design, including datasets, performance metrics, and case studies evaluating the method effectiveness, comparative performance, etc. The paper is concluded in Section V.

II. RELATED METHODOLOGIES

In this section, we provide an overview of the fundamentals of the Tree-structured Parzen Estimator (TPE) and eXtreme Gradient Boosting (XGBoost), which are critical components of the proposed LARSI-TPE-XGB model. These methods are designed to efficiently handle high-dimensional data and adapt effectively, thereby enhancing load forecasting accuracy.

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TABLE II
MAIN DISTINGUISHING FEATURES OF THE PROPOSED SCHEME COMPARED TO SIMILAR METHODS

Method	Data Requirement	Feature Engineering	Hyperparameter Tuning	Time-based Update/Adaption	Highlights
Traditional machine learning (e.g., [27]-[30])	Low to moderate	Manual (calendar data, historical statistical data, and optional weather data)	Mostly manual or default tuning	Rarely considered	Requires substantial manual feature engineering and lacks adaptability.
Deep Learning (e.g., [31]-[32])	High (large datasets)	Typically uses only load history and calendar data, with optional weather inputs.	Expert-designed networks/parameters	Rarely considered	Automatically extracts complex features but relies on data availability, with data more likely to be lacking, especially for coarse-grained time series [28].
LARSI-TPE-XGB (Evolved from traditional machine learning)	Low to moderate	Manual (calendar data, historical statistical data, and optional weather data) + **the proposed novel LARSI**	Automated & dynamic	Considered	Proposes novel LARSI for feature enhancement in load forecasting. Handles high-dimensional data, and updates hyperparameters dynamically using TPE-XGB.

A. Extreme Gradient Boosting

eXtreme Gradient Boosting (XGBoost) is a type of gradient-boosting algorithm [16], [24]. This versatile algorithm has applications across various domains. For example, a flexible XGBoost model for forecasting power generation was proposed [17], demonstrating its utility in the energy sector. In the field of epidemiology, XGBoost was leveraged to predict trends in the spread of COVID-19 infections based on key features [33]. These applications exemplify the broad applicability of XGBoost across various fields, particularly for tasks that require strong feature selection capabilities. When combined with other technologies, XGBoost can enhance their effects.

The core of XGBoost is an ensemble of decision trees. To prevent overfitting and to accelerate the training process, XGBoost employs randomization techniques and incorporates a sparsity-aware algorithm for effectively handling missing values. Let M denote the total number of trees, and let x_i be the feature vector of the i-th data sample. We denote by $f_m(x_i)$ the prediction of the m-th tree on x_i. Hence, the ensemble model's prediction ($\hat{y}_i$) for the i-th sample is given by [34]:

$$\hat{y}_i = \sum_{m=1}^{M} f_m\left(x_i\right), \quad f_m \in F \tag{1}$$

where f_m denotes an individual decision tree, and F is the space of all possible trees.

XGBoost uses an additive model where trees are added one at a time. The objective function for all M trees is:

$$\begin{aligned} \text{Obj}\left(\theta\right) &= \sum_{i=1}^{n} \mathcal{L}\left(y_i, \hat{y}_i\right) + \sum_{m=1}^{M} \Omega\left(f_m\right) \\ \theta &= \left(f_1,\ f_2,\ \ldots,\ f_M\right) \end{aligned} \tag{2}$$

where n represents number of training sample; $\mathcal{L}(y_i, \hat{y}_i)$ is a differentiable loss function; $\Omega(f_m)$ is a regularization term penalizing the complexity of tree f_m.

In XGBoost, the regularization term $\Omega(f)$ for a tree f with T leaves and leaf weights $\{\omega_j\}_{j=1}^{T}$ is typically written as:

$$\Omega\ \left(f\right) = \zeta T + \frac{1}{2}\lambda \sum_{j=1}^{T} \omega_j^2 \tag{3}$$

where ζ denotes penalty parameter for the number of leaves. λ represents L2 penalty parameter for leaf weights.

To simplify the complexity of training M trees, XGBoost employs a forward stage-wise approach. At each iteration t, a new tree f_t is added to adjust for the residuals based on the current model prediction. Specifically, at the t-th step, the prediction for x_i becomes $\hat{y}_i^t = \hat{y}_i^{t-1} + f_t(x_i)$. Hence, the objective function at step t is:

$$\begin{aligned} Obj(\theta)^t &= \sum_{i=1}^{n} \mathcal{L}\left(y_i,\ \hat{y}_i^t\right) + \sum_{m=1}^{t} \Omega\left(f_m\right) \\ &= \sum_{i=1}^{n} \mathcal{L}\left(y_i,\ \hat{y}_i^{t-1} + f_t\left(x_i\right)\right) + \sum_{m=1}^{t} \Omega\left(f_m\right) \end{aligned} \tag{4}$$

In each iteration, f_t is chosen to minimize the incremental loss plus regularization, leading to a new leaf structure and updated leaf weights. Through this stage-wise construction, XGBoost effectively balances model complexity with data fidelity while leveraging advanced techniques to handle large-scale and possibly sparse datasets.

B. Tree-Structured Parzen Estimator

Tree-structured Parzen Estimator (TPE) [23] is utilized for hyperparameter optimization. This approach is particularly relevant in the context of power load prediction, a task characterized by its time-series basis and greater levels of noise than other similar tasks. TPE is a technique within the domain of sequential model-based optimization (SMBO) designed to identify hyperparameters that yield reduced loss values. It strategically navigates through the hyperparameter space to identify regions with the potential for lower losses.

In the TPE algorithm, the function $l(\phi)$ denotes the density distribution based on the observed set $\{\phi(i)\}$ where the associated loss $f(\phi(i))$ falls below a threshold τ^*. Conversely, $g(\phi)$ signifies the density distribution derived from the rest of the observed data points.

$$p\ \left(\phi|\tau\right) = \begin{cases} l\left(\phi\right),\ if\ \tau < \tau^* \\ g\left(\phi\right),\ if\ \tau \geq \tau^* \end{cases} \tag{5}$$

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In optimization, the choice of parametrizing $p(\phi, \tau)$ as $p(\tau)p(\phi|\tau)$ in the TPE algorithm was made to simplify the optimization process of expected improvement (EI).

$$EI_{\tau*}(\phi) = \int_{-\infty}^{\tau^*} (\tau^* - \tau)\, p(\phi|\tau)\, p(\tau)\, d\tau = \int_{-\infty}^{\tau^*} \frac{(\tau^* - \tau)\,(p(\phi|\tau)\, p(\tau))}{p(\phi)} d\tau \tag{6}$$

As derived, $\gamma = p(\tau < \tau^*)$ and $p(x) = \int_{\mathbb{R}} p(\phi|\tau)p(\tau)d\tau = \gamma l(\phi) + (1-\gamma)g(\phi)$.

Consequently,

$$\begin{aligned} &\int_{-\infty}^{\tau^*} (\tau^* - \tau)\, p(\phi|\tau)\, p(\tau)\, d\tau \\ &= l(\phi) \int_{-\infty}^{\tau^*} (\tau^* - \tau)\, p(\tau)\, d\tau \\ &= \gamma\tau^* l(\phi) - l(\phi) \int_{-\infty}^{\tau^*} p(\tau)\, d\tau \end{aligned} \tag{7}$$

Finally,

$$\begin{aligned} EI_{\tau*}(\phi) &= \frac{\gamma\tau^* l(\phi) - l(\phi) \int_{-\infty}^{\tau^*} p(\tau)\, d\tau}{\gamma l(\phi) + (1-\gamma)\, g(\phi)} \\ &\propto \left(\gamma + \frac{g(\phi)}{l(\phi)}(1-\gamma)\right)^{-1} \end{aligned} \tag{8}$$

The final statement demonstrates that when attempting to maximize improvement, our objective is to identify points x that exhibit a high possibility according to $l(\phi)$ while simultaneously having a low possibility according to $g(\phi)$. The tree-like structure of l and g simplifies the process of generating numerous candidates based on l and assessing them by the ratio $g(\phi)/l(\phi)$. During each iteration, the algorithm produces the candidate ϕ^* that yields the EI.

The execution of the TPE algorithm involves:

1) Randomly selecting several sets of hyperparameters as initial observations.
2) Repeating the following process until a predefined stopping criterion is met:
   a. Evaluating the performance of sampled sets using a specific statistical measure to determine τ^*.
   b. Choosing a new set of hyperparameters and applying the EI criterion, focusing on maximizing the ratio $l(\phi)/g(\phi)$ to identify the optimal set ϕ^*.
3) Finally, the algorithm selects the best-performing hyperparameter combination as the optimization result.

III. Proposed Methodology

This section delves into the methodology behind the proposed LARSI-TPE-XGB by providing an overview of the proposed load forecasting approach and including a flowchart, the feature extraction process, and the generation and incorporation of the proposed LARSI features. In addition, the synergy between LARSI, TPE, and XGBoost in enhancing load forecasting accuracy is described in detail.

A. LARSI-TPE-XGB Method

The flowchart of the proposed LARSI-TPE-XGB method for load forecasting is shown in Fig. 1 and consists of the following steps:

First, the load data is prepared. Then, data preprocessing is performed, including handling missing values through linear interpolation and converting time units to daily granularity. The next step is feature extraction, which is essential for machine learning to derive useful information from raw data. In other studies, basic, calendar-based, and weather feature data are often extracted, as demonstrated in [35]. However, this study focuses on enhancing STLF. Therefore, we propose the Load Adaptive Relative Strength Index (LARSI). The Relative Strength Index (RSI), developed by J. Welles Wilder Jr., is a widely used technical indicator for short-term stock price forecasting. Recognizing the similarities between stock markets and power systems, this paper proposes LARSI for use in power systems as a feature to enhance STLF.

The processed data is then continuously partitioned into training, validation, and test sets to reflect continuous updates over time. As the data evolves, both the data and its hyperparameters must be updated using an expanding window or rolling window approach, as shown in Fig. 2. Next, our proposed model, TPE-XGB, is employed to forecast daily average load. Since the size of time series datasets is usually limited, especially in coarse-grained forecasting, feature extraction becomes particularly important. We chose XGBoost as the prediction algorithm because it can automatically select appropriate high-dimensional features, effectively prevent overfitting, and enhance the model's generalization capability. Additionally, XGBoost is especially effective when combined with our proposed LARSI. Tree-structured Parzen Estimator (TPE) enables continuous model optimization, allowing automatic parameter tuning and updating as new data becomes available, ensuring the continued relevance of machine learning models over time.

B. Feature Extraction of Basic and Calendar-Based Features for Short-Term Load Forecasting

The process of feature extraction involves identifying and extracting relevant information, referred to as features, from raw data. These extracted features simplify the data analysis and modeling processes, thereby enhancing their effectiveness for predictive modeling and other analytical tasks.

In this subsection, we focus on the extraction of features from time-series data. We adhere to basic (baseline and moving) and calendar-based feature extraction guidelines, as shown in Table III, which include the following three parts:

1) Baseline features: Previous load values serve as key indicators, providing insights based on historical data. For example, the load records from one week ago can be significantly related to the target load value.

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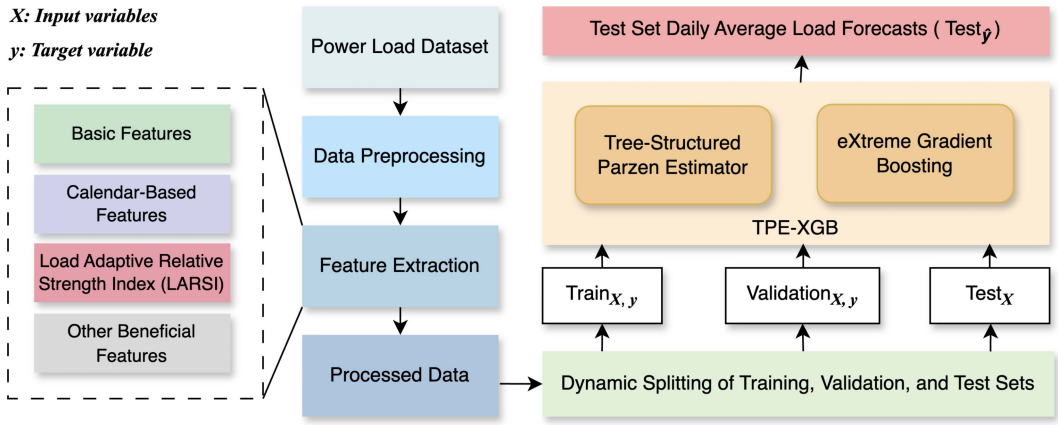


Fig. 1. Flowchart of the proposed LARSI-TPE-XGB for load forecasting.

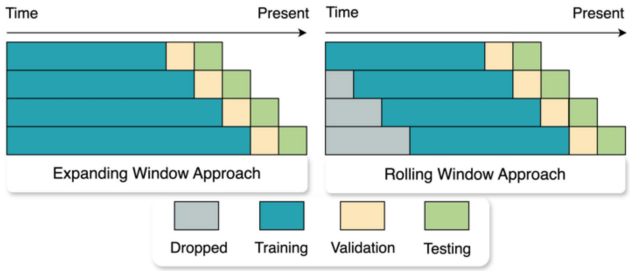


Fig. 2. Illustration of the rolling and expanding window approaches.

TABLE III
THE BASIC AND CALENDAR-BASED FEATURES

Feature Name	N Range	Definition
Baseline Features		
Historical Load	1~30	The load measurement from N days prior.
Moving Features		
Moving SVD	2~30	The standard deviation of the load from one day prior up to N days in the past.
Moving MIN	2~30	The lowest load value recorded from one day prior up to N days in the past.
Moving MAX	2~30	The highest load value recorded from one day prior up to N days in the past.
Moving AVG	2~30	The mean load value calculated from one day prior up to N days in the past.
Calendar Features		
Day of Week Week of Year Month of Year Quarter of Year Non-Working Day		

2) Moving features: Considering the maximum, minimum, standard deviation, and average loads over recent days enables the model to better capture patterns and fluctuations within the time series.
3) Calendar features: Calendar effects can be represented using a polar coordinate system, as shown in Equation (9) [36]. The non-working day attribute is set as 1 for national holidays or weekends, and otherwise set as 0 (zero).

$$z = \begin{bmatrix} \cos(g) \\ \sin(g) \end{bmatrix}, g = \frac{2\pi t}{p} \tag{9}$$

where t denotes time and p indicates the length of cycle.

In addition, some studies have employed numerical weather prediction (NWP) as input features. However, in this study, we focus primarily on enhancing STLF using the proposed LARSI method. Due to the difficulty in finding long-term datasets that include NWP data, we employed two larger electricity load datasets without NWP and one smaller dataset with temperature information for experimentation, as detailed in Section IV.

The feature matrix X, following feature extraction, can be represented as $[x_1, x_2, \ldots, x_n]^{\mathrm{T}}$, where x_i is a vector containing the extracted features. The target variable $y = [y_1, y_2, \ldots, y_n]^{\mathrm{T}}$, where y_i corresponds to the actual load at time point i. The complete mathematical expression can be represented as:

$$[X|y] = \left[\begin{array}{cccc|c} x_{11} & x_{12} & \cdots & x_{1m} & y_1 \\ x_{21} & x_{22} & \cdots & x_{2m} & y_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{nm} & y_n \end{array}\right] \tag{10}$$

where n is the number of days, and m is the number of features.

In addition, all features inputted into the model have undergone standardization, which aims to scale all features to a consistent scale. The formula for standardization is defined as:

$$x_{standardized} = \frac{x_{orig} - \mu}{\sigma} \tag{11}$$

where x_{orig} represents the original feature value; μ and σ denote the mean and standard deviation of the training set, respectively. After standardization based on the training set to prevent data leakage, each feature has a mean of 0 and a standard deviation of 1.

C. Proposed LARSI for Enhanced Load Forecasting

To overcome the limitation of additional features developed beyond those mentioned in Section III-B, we propose a novel feature, the Load-Adaptive Relative Strength Index (LARSI),

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for load forecasting. This innovation is based on the Relative Strength Index (RSI) [37], a widely recognized technical indicator in the stock market primarily used for short-term price prediction. Given the similarities between the stock market and power systems as complex nonlinear systems that both benefit from short-term forecasting, we propose integrating RSI into STLF. This approach is not merely a straightforward transfer of financial techniques to power systems; instead, it leverages the similar characteristics of these systems. We have improved the standard RSI to develop LARSI for load forecasting, providing a novel approach to existing forecasting features.

Daily loads are more stable than hourly loads because they mitigate short-term fluctuations and reduce the accumulation of errors due to noise from sudden, uncontrollable factors. Daily loads offer many benefits for power systems, as described in Section I. Some studies have used technical indicators, including the standard RSI, to predict next-day stock prices [38], similar to daily load forecasting.

Incorporating RSI into the stock market often involves using a simple moving average (SMA) for a specific period, commonly 14. For n in RSI_{SMA}, the formula is as follows:

$$RSI_{SMA}\left(n\right) = 100 - \frac{100}{1 + RS\left(n\right)} \tag{12}$$

where:

$$RS_{SMA}\ \left(n\right) = \frac{n\ days\ avg.\ increased}{n\ days\ avg.\ decreased} \tag{13}$$

The RSI spans a scale of 0 to 100. Values over 80 suggest that the asset is overbought, whereas values under 20 suggest that the asset is oversold [38].

However, when applying RSI to load forecasting, certain limitations must be addressed. Fixed thresholds may not adapt well to varying load conditions, resulting in inaccuracies, or a single RSI may fail to capture changes and trends across different time periods. Therefore, it is necessary to employ machine learning to simultaneously consider multiple RSI features for more precise regression analysis, rather than relying on simple threshold judgments.

Building on the above foundation, we further propose considering RSI_{EMA}, the exponential moving average (EMA) variant of RSI, which places more emphasis on recent loads. Therefore, RSI_{EMA} is more sensitive to recent load changes. To calculate RSI_{EMA}, we first need to determine:

$$\Delta Load_t = Load_t - Load_{t-1} \tag{14}$$

Then, separate the positive changes and negative changes into Up and Down by the following two conditions:

$$Up_t = \begin{cases} \Delta Load_t,\ if\ \Delta Load_t > 0 \\ 0,\ else \end{cases} \tag{15}$$

$$Down_t = \begin{cases} \Delta Load_t,\ if\ \Delta Load_t < 0 \\ 0,\ else \end{cases} \tag{16}$$

The formula for calculating the smoothing factor α is as follows. The larger the α, the more sensitive the EMA is to recent load changes; the smaller the value, the smoother the EMA.

$$\alpha\ = \frac{2.0}{period + 1} \tag{17}$$

We can then calculate the EMA for Up and Down using the following formulas.

$$Up_{EMA,t} = (1 - \alpha)\ * Up_{EMA,t-1} + \alpha * Up_t \tag{18}$$

$$Down_{EMA,t} = (1 - \alpha)\ * Down_{EMA,t-1} + \alpha * Down_t \tag{19}$$

The formula for RSI_{EMA} can be obtained by:

$$RSI_{EMA},_t = 100 - \frac{100}{\left(1 + \frac{Up_{EMA,t}}{Down_{EMA,t}}\right)} \tag{20}$$

In addition, according to [39], applying smoothing and denoising techniques to the original sequence can reduce errors in time series forecasting. Therefore, the Savitzky-Golay (SG) filter was employed to smooth and denoise the data prior to calculating the RSI. This filtering method preserves both the shape and width of the data as it eliminates noise. A subsequence of X, using a window size defined by $n\ = 2m + 1$, can be expressed as follows:

$$\left\{x_{s-m}, \ldots, x_s, \ldots, x_{s+m}\right\},\ s \in [m+1, T-m] \tag{21}$$

A polynomial $p(i)$ of order R is used to fit data points within the window and is expressed as:

$$p\ \left(i\right) = \sum_{v=0}^{R} \alpha_v i^v\ ,\ i \in [-m, m] \tag{22}$$

where α_v represents the v-th coefficient of the filter.

Then, the least-square method is used to minimize the following error:

$$\varepsilon\ = \sum_{i=-m}^{m} \left(p\left(i\right) - x_{s+i}\right)^2 \tag{23}$$

Calculating a_0 yields $p(0)$, the best fit for the central point x_s. By sliding the window, each point in X becomes a center point. This process generates the smoothed sequence $\bar{X}$.

The RSI effectively indicates the rate of change in electricity demand: higher values indicate rapid increases in demand, whereas lower values reflect decreases. This insight aids in understanding the evolution of demand. Furthermore, RSI aids in identifying potential demand extremes, thus improving forecasts of electricity demand.

To optimize STLF, we employ RSI values calculated using periods n ranging from 2 to 30 days (approximately one month) for each calculation method previously mentioned. Fig. 3 provides a comprehensive spectrum of RSI feature calculations. We call this combination of RSI calculations LARSI. In addition, to handle such high-dimensional features, it is essential to consider feature selection, hyperparameter tuning, and algorithm choice, which are described in the following subsection.

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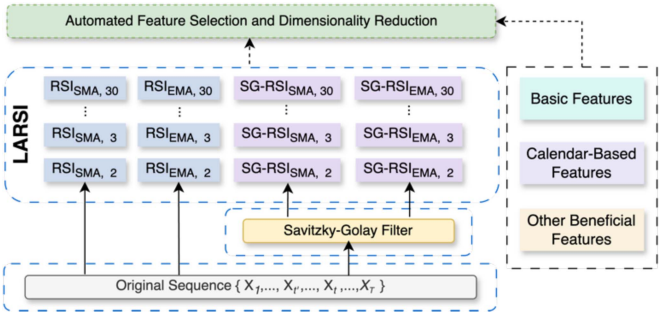


Fig. 3. Process of generating the proposed LARSI.

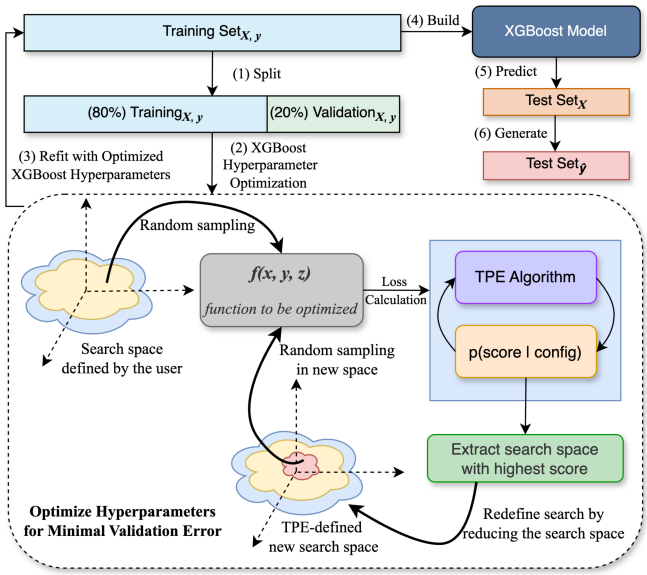


Fig. 4. Flowchart of the TPE-XGB model building and forecasting process.

D. Synergizing LARSI With TPE and XGBoost

The ability to work with high-dimensional features is a vital part of effective load forecasting. We propose using the XGBoost algorithm [24], which can effectively handle high-dimensional data, automatically identify the most relevant features, and assign appropriate weights [33] as mentioned in Section II-A. By determining the interactions and impacts between multiple features, XGBoost can discover complex patterns and relationships that are difficult for humans to detect.

However, there are challenges involved in applying machine learning to load forecasting. Hyperparameters significantly impact the performance of the forecasting model, especially in high-dimensional scenarios, rendering XGBoost particularly sensitive to hyperparameter selection. To achieve optimal performance, careful model tuning is imperative. Due to the numerous hyperparameters used by XGBoost, such as learning rate, maximum depth, subsample ratio, regularization parameters, etc., tuning the model is a highly challenging task [40]. Moreover, for any time series predictions, load forecasting requires dynamic updates of data and model parameters over time to ensure that recent trends are captured [41], as illustrated by the concept in Fig. 2. Consequently, given that machine learning models are sensitive to hyperparameter settings, the optimal hyperparameter selection may change as time goes on, thereby affecting predictive performance.

For these reasons, this study proposes the use of the Tree-structured Parzen Estimator (TPE) for automatic hyperparameter optimization in the XGBoost model. The theoretical foundation of TPE is detailed in Section II-B. By combining XGBoost's capability to handle high-dimensional features with TPE's efficiency in hyperparameter tuning, this approach aims to enhance both feature selection and forecasting, ultimately improving load forecasting performance. Since XGBoost inherently performs feature selection during training, and its built-in feature selection mechanism, as well as TPE's assistance in hyperparameter tuning, can effectively mitigate the risk of overfitting, it may not be necessary to adjust feature-related parameters. We refer to the resulting process as TPE-XGB.

The proposed process is shown in Fig. 4 and the detailed optimization procedure can be found in [36]. In this process, the last 20% of the training set is first allocated to the validation set. Next, TPE is used for optimization, finding the set of hyperparameters that yields the least errors in the validation set. Subsequently, the training set is refitted to build an XGBoost model. The model simultaneously captures the most recent trends, as the training and validation sets update over time, as shown in Fig. 2. Finally, the established XGBoost model is used to predict future loads. This approach not only ensures that the model captures recent trends, but also employs optimized hyperparameters.

IV. Simulation Studies and Analysis

To demonstrate the feasibility and effectiveness of the proposed method, we present in this section the real-world public datasets used in the simulation studies, as well as the commonly employed performance metrics and statistical analysis methods for load forecasting. We designed a series of simulation experiments to validate our approach.

The design of the simulation studies comprises four distinct cases to thoroughly evaluate the proposed method. Case I is dedicated to assessing the effectiveness of the proposed LARSI-TPE-XGB. Case II addresses the feature importance analysis of LARSI and the statistical validation of LARSI-TPE-XGB. In Case III, we compare our approach with benchmark methods, employing the current state-of-the-art load forecasting models. Case IV investigates the impact of weather features on load forecasting accuracy. In these experiments, we update the time-series data annually, with TPE iterations set to 50. Our goal is to predict the average load value on day t, primarily based on information from days $t-1$ to $t-n$ (where $n=30$), generating features as described in Section III. Due to the distinct characteristics, variations in trends, and seasonality between the load datasets, individual models were trained separately for each dataset. Furthermore, to predict the future data of the same dataset without introducing look-ahead bias, an expanding window (Fig. 2) data updating strategy was employed.

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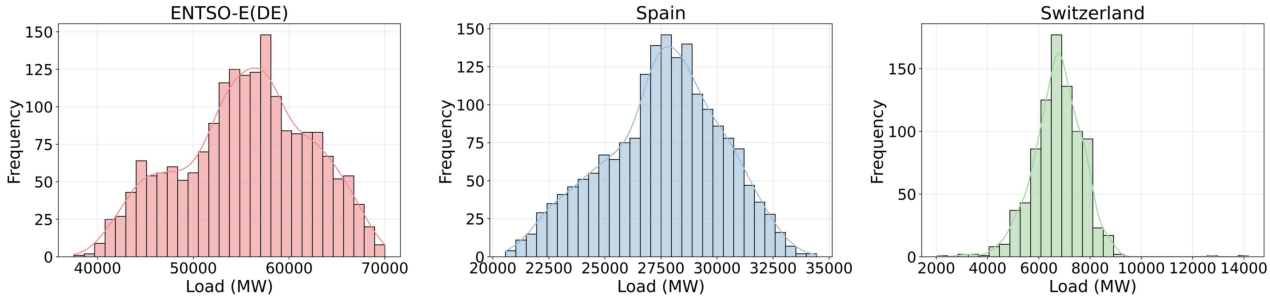


Fig. 5. Daily average load distributions for the ENTSO-E (DE), Spain, and Switzerland datasets.

TABLE IV
SUMMARY OF TRAINING AND TESTING TIME PERIODS

Dataset	Train		Test	
	Start	End	Start	End
ENTSO-E(DE)	01-01-2019	12-31-2022	01-01-2023	12-31-2023
	01-01-2019	12-31-2023	01-01-2024	03-31-2024
Spain	01-01-2014	12-31-2016	01-01-2017	12-31-2017
	01-01-2014	12-31-2017	01-01-2018	12-31-2018
Switzerland	01-31-2015	06-30-2016	07-01-2016	04-15-2017

Note: Dates are formatted as MM-DD-YYYY.

A. Datasets

To verify the effectiveness of the proposed method across different datasets, we assessed the performance of our models on the following real-world public electricity load datasets:

1) ENTSO-E(DE): Data from Germany spanning from January 2019 to March 2024 [42]. It incorporates recent data, ensuring the proposed method's effectiveness with up-to-date information.
2) Spain: Data from Spain from 2014 to 2018 [43]. This dataset serves to demonstrate the applicability of the proposed method in a different geographical context.
3) Switzerland: Data from Switzerland spanning from Jan. 2015 to May 2017 [44]. Although this dataset is limited, it includes weather features such as temperature.

We employed the ENTSO-E(DE) and Spain datasets to evaluate the performance of LARSI and TPE-XGB, while the Switzerland dataset was employed to conduct feature analysis, focusing on LARSI and weather-related factors.

Fig. 5 illustrates the daily average load distribution for the datasets. Table IV provides an overview of the training and testing periods of the datasets, which are split using an expanding window approach for the ENTSO-E(DE) and Spain datasets. The last 20% of the training set is used as a validation set, as detailed in Fig. 4. The data are derived from the features presented in Table III and Fig. 3 to generate the matrix in the format of (10), which is then used to build the model.

B. Evaluation Metrics

In this study, we employed three well-established evaluation metrics to evaluate the accuracy of load forecasting: mean absolute error (MAE), root mean squared error (RMSE), and mean absolute percentage error (MAPE). The specific definitions of these metrics are as follows:

$$MAE = \frac{1}{T}\sum_{i=1}^{T}\left|f_p(t) - f_o(t)\right| \tag{24}$$

$$RMSE = \sqrt{\frac{1}{T}\sum_{i=1}^{T}\left(f_p(t) - f_o(t)\right)^2} \tag{25}$$

$$MAPE = \frac{100\%}{T}\sum_{i=1}^{T}\left|\frac{f_p(t) - f_o(t)}{f_o(t)}\right| \tag{26}$$

where $f_p(t)$ and $f_o(t)$ represent the predicted load and observed load at time t, respectively. T denotes the total number of predictions.

In general, smaller MAE and RMSE values reflect better model performance. MAE quantifies the average magnitude of errors, regardless of error direction. RMSE places more weight on larger errors, thus making it more sensitive to outliers.

However, using traditional metrics without conducting statistical significance tests can lead to erroneous conclusions in the evaluation of model performance. To address this issue, the DM test [20] was developed to assess whether there are significant differences in predictive accuracy between two models. In this test, assume that models A and B make predictions over a certain period, producing prediction error sequences $e_A = (a_1, a_2, \ldots, a_T)$ and $e_B = (b_1, b_2, \ldots, b_T)$, respectively. Then, calculate the error difference sequence $d = (d_1, d_2, \ldots, d_T)$, where $d_i = a_i - b_i$ for $i = 1, 2, \ldots, T$. The DM statistic is calculated as follows [45]:

$$DM = \frac{\sqrt{T} * d_{mean}}{d_{std}} \tag{27}$$

where d_{std} indicates the standard deviation of vector d; T denotes the sample size, and d_{mean} represents the mean of vector d.

The DM test can be interpreted as follows: A DM value of zero (DM = 0) indicates no statistically significant difference in predictive performance between the two models. A DM value less than 0 suggests that Model A outperforms Model B and values greater than zero suggests that Model B outperforms Model A.

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TABLE V
FORECASTING ERRORS OF DISTINCT CONFIGURATIONS

Dataset	Model	MAE	RMSE	MAPE
ENTSO-E(DE)	Basic-XGB	1387.64	1781.00	2.73%
	LARSI-XGB	1272.22	1652.43	2.52%
	LARSI-TPE-XGB	**1096.32**	**1450.00**	**2.15%**
Spain	Basic-XGB	846.78	1361.85	3.00%
	LARSI-XGB	808.40	1263.71	2.85%
	LARSI-TPE-XGB	**640.25**	**943.24**	**2.23%**

Note: MAE and RMSE values are in **MW**.

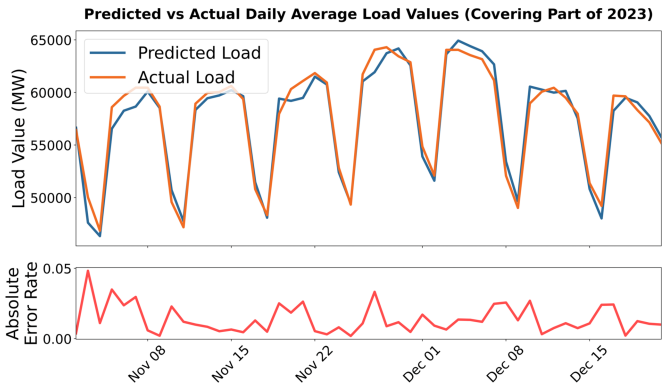


Fig. 6. Comparison of predicted and actual daily average load values from a portion of the test set of the ENTSO-E (DE) dataset.

C. Case I: Evaluating the Effectiveness of the Proposed LARSI-TPE-XGB

To demonstrate the effectiveness of the proposed LARSI and TPE-XGB, we conducted tests on the ENTSO-E(DE) and Spain datasets. In these experiments, we used XGBoost and divided our experimental evaluation into three distinct configurations: (i) Basic-XGB used the features listed in Table III; (ii) LARSI-XGB combined the features from Table III with the LARSI features; (iii) LARSI-TPE-XGB was used as an extension of LARSI-XGB, applying the TPE algorithm for hyperparameter optimization. The hyperparameters of Basic-XGB and LARSI-XGB are set to their default values, which are primarily established based on XGBoost's empirical design principles and practical experience.

The forecasting errors are presented in Table V. First, we compare Basic-XGB and LARSI-XGB; the RMSE improved by 7.20%, and MAPE by 0.15%−0.21% for both datasets when using LARSI. These results suggest that the proposed LARSI can enhance load forecasting accuracy. Additionally, when comparing LARSI-XGB and LARSI-TPE-XGB, the MAE and RMSE improved by approximately 13% in the ENTSO-E(DE) dataset. In the Spain dataset, the MAE and RMSE improved by 20.83% and 25.35%, respectively. This indicates that hyperparameter tuning is critical when using XGBoost in such high-dimensional contexts, validating our hypothesis.

The predicted versus actual daily average load values for the portion of the test set of the ENTSO-E (DE) dataset, covering November to December 2023, as shown in Fig. 6, demonstrate that most predicted load values align closely with the actual values. This is evidenced by a MAPE of only 2.15% on the test set. Additionally, the figure reveals a few instances where the error exceeds 3–5%, which is expected due to the inherent uncertainties in forecasting future loads.

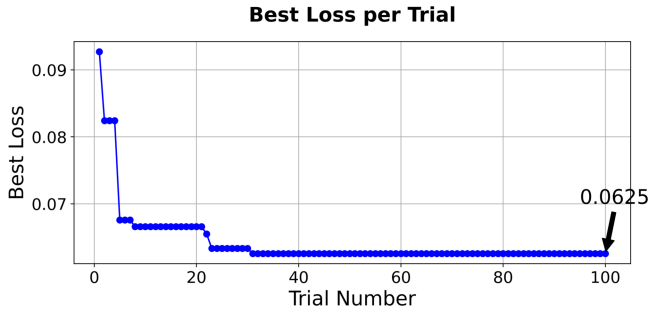


Fig. 7. The convergence of TPE-XGB to the best loss.

To clarify the convergence characteristics of the LARSI-TPE-XGB, we utilized HyperOpt-Sklearn [46] to optimize the XGBoost implementation. The output from this framework enabled us to visualize the optimal loss trajectory, as shown in Fig. 7. The loss metric is defined as $1 - R^2$. During one of the expanding window iterations, the model achieved a MAPE of 2.13% and a loss of 0.054 on the test set. A comparative analysis between the optimal loss observed in Fig. 7(0.0625) and the test set loss (0.054) demonstrates a strong alignment, with a slightly lower test error. This consistency strongly supports the robust generalization ability, highlighting its effectiveness in forecasting unknown future data. Furthermore, this provides empirical evidence supporting the efficacy of the proposed method and hyperparameter optimization methodologies for load forecasting.

Although the MAE and RMSE values obtained from certain datasets—such as the ENTSO-E(DE) dataset—appeared rather high, this was primarily because the load scale in that dataset ranged from tens of thousands to hundreds of thousands of megawatts (as shown in Fig. 5). Thus, a seemingly large absolute error was actually normal under these circumstances. More importantly, the experimental results indicated that the MAPE for this dataset was 2.15%, suggesting that even for a large-scale power system, the actual prediction error ratio remained around 2%. This level of relative error was sufficiently accurate and feasible for various practical applications—such as power trading strategies, cross-border scheduling, improving transmission efficiency, estimating fuel demand for thermal power plants, etc.—as mentioned in Section I. Moreover, due to the distinct characteristics, variations in trends, and seasonality among the load datasets, we used three load datasets to evaluate our proposed method, rather than relying on the error from only one dataset. In addition, Case III provides a comparison with other forecasting methods to ensure a more comprehensive evaluation.

D. Case II: Feature Importance Analysis and Statistical Validation of the Proposed LARSI-TPE-XGB Model

To provide a more comprehensive analysis, we present the model's feature importance, as illustrated in Fig. 8. Since each feature accounts for multiple time periods, we computed them

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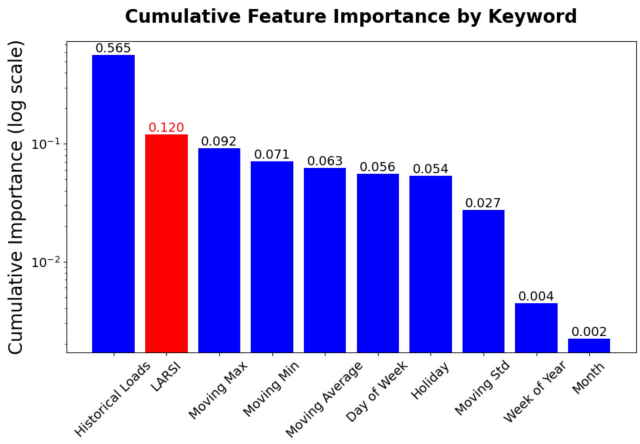


Fig. 8. Cumulative feature importance.

TABLE VI
DM TEST USING DIFFERENT METHODS IN THE ENTSO-E(DE) AND SPAIN DATASETS

Dataset	Methods	DM statistic	*p*-value
ENTSO-E(DE)	LARSI-XGB *versus* Basic-XGB	-2.409	1.639e-02
	LARSI-TPE-XGB *versus* LARSI-XGB	-4.628	4.819e-06
Spain	LARSI-XGB *versus* Basic-XGB	-3.124	1.854e-03
	LARSI-TPE-XGB *versus* LARSI-XGB	-6.648	5.804e-11

Note: 5% significance level.

cumulatively. We observed that the most important feature was recent past loads (historical data), which is reasonable. Unexpectedly, the proposed LARSI ranked second, surpassing some moving and calendar-based features, further corroborating the effectiveness of LARSI in load forecasting.

To further analyze LARSI and TPE-XGB, we employed the DM test to compare errors. The DM test, which compares errors and provides statistical significance, is well-suited for time series data. A DM value below zero indicates model A outperforms model B. A *p*-value under 5% suggests statistical significance sufficient to reject the hypothesis of equal predictive performance between methods. The results in Table VI indicate that LARSI-XGB significantly outperforms Basic-XGB, while LARSI-TPE-XGB further enhances performance over LARSI-XGB in both datasets. These results confidently demonstrate the effectiveness of LARSI and TPE-XGB in load forecasting.

E. Case III: Comparison With Current Leading Methods

We compared the accuracy of LARSI-TPE-XGB against current leading models using the DM test. As the application of deep learning (DL) in load forecasting increases, the potential of DL for feature extraction and pattern recognition has been recognized. For example, recurrent LSTM neural networks have been utilized for load forecasting [19], while the combination of CNN and GRU models has achieved notable success in this domain [47]. These approaches typically rely on raw features extracted from historical load data using DL techniques. TCN networks have further demonstrated the capability of DL in analyzing historical load data across multiple timescales [48].

To showcase the advantages of the proposed method, we benchmarked it against the latest and most advanced technologies in the field. In the comparative experiment, the inputs were categorized into three types to ensure a fairer and more comprehensive analysis:

1) *Deep learning:* We used the past 30 days of load data, combined with the calendar features outlined in Table III, as inputs for a deep learning model following the methodologies described in [18], [19], [47], [49], [50], and [51].
2) *Shallow learning:* To ensure consistency and objectivity in our comparison, we adopted the features from those listed in Table III and the proposed LARSI. For a more accurate evaluation, some of the models also utilized TPE.
3) *Shallow learning (2):* Similar to (ii), with LARSI replaced by the standard RSI to demonstrate the improvements achieved by the proposed LARSI.

As the ENTSO-E(DE) dataset is more recent, we used it as the basis for comparative analysis. The benchmark models considered were as follows:

1) Seasonal Naive Method—A simple baseline forecasting method that uses the load value from one week ago as the prediction.
2) CNN-GRU—Convolutional neural network combined with gated recurrent unit [47];
3) CNN-LSTM—Convolutional neural network combined with long short-term memory [47];
4) CNN-LSTM (2)—Another variant of CNN-LSTM architecture [49];
5) TCN-MLP—Temporal convolutional network combined with multi-layer perceptron [50];
6) LSTM—Long short-term memory network [19];
7) Bi-LSTM—Bidirectional long short-term memory network [51];
8) ReluTCN—Temporal convolutional network with ReLU activation [18], [48];
9) kNN—k-nearest neighbors algorithm;
10) RF—Random forest algorithm;
11) LGBM—Light gradient-boosting machine based on decision trees [52];
12) TPE-kNN—Tree-structured Parzen Estimator optimized k-nearest neighbors;
13) TPE-RF—Tree-structured Parzen Estimator optimized random forest;
14) TPE-LGBM—Tree-structured Parzen Estimator optimized light gradient boosting machine;
15) TPE-XGB (STDRSI)—Tree-structured Parzen Estimator optimized XGBoost with Standard RSI instead of LARSI.

Table VII presents the DM statistics and corresponding *p*-values for the benchmark models compared to our proposed LARSI-TPE-XGB, where "No" indicates the model number. A higher DM value indicates superior model performance compared to the LARSI-TPE-XGB, and a *p*-value below 0.05 indicates statistical significance. The results reveal that shallow

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TABLE VII
DM STATISTICS AND P-VALUES OF BENCHMARK MODELS COMPARED TO THE PROPOSED LARSI-TPE-XGB

No	Input type	DM statistic	p-value	No	Input type	DM statistic	p-value
1		-6.396	4.03e-10	9	(ii)	-5.655	2.75e-08
2	(i)	-7.355	9.02e-13	10	(ii)	-4.142	4.11e-05
3	(i)	-4.153	3.91e-05	11	(ii)	-2.735	6.47e-03
4	(i)	-9.784	0.00e+00	12	(ii)	-5.043	6.64e-07
5	(i)	-9.630	0.00e+00	13	(ii)	-4.064	5.66e-05
6	(i)	-6.148	1.72e-09	14	(ii)	-2.159	3.13e-02
7	(i)	-7.148	3.53e-12	15	(iii)	-2.135	3.33e-02
8	(i)	-9.350	0.00e+00	Note: 5% significance level.			

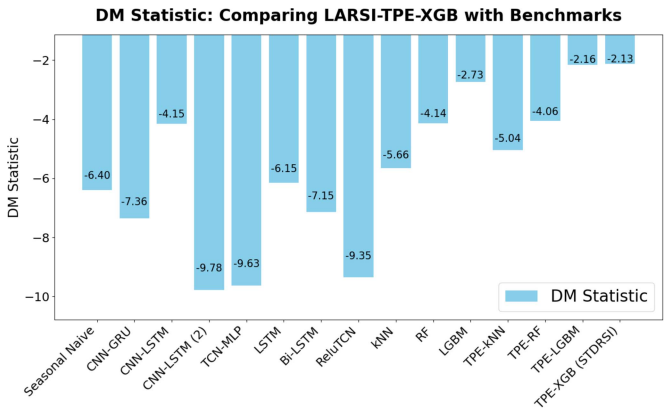


Fig. 9. DM statistics of benchmark models in comparison to the proposed LARSI-TPE-XGB.

learning models outperform deep learning models in most cases within this context. In fact, some deep learning methods are even inferior to the seasonal naive method. This outcome can be explained by the limited available data, particularly in daily forecasting scenarios where careful selection of predefined features is an important factor, such as the proposed input (ii). This finding highlights the rationale behind the development of LARSI.

In our analysis, the best-performing deep learning model was CNN-LSTM. Despite its lack of reliance on predefined time series features, CNN-LSTM effectively demonstrated its capability to automatically extract relevant patterns from the data, achieving performance levels comparable to the RF model enhanced by LARSI. We compared TPE-LGBM, which yielded promising results, evidenced by a DM value of -2.159. Fig. 9 shows the DM statistics of benchmark models in comparison to the proposed LARSI-TPE-XGB. If this value were within the range of -1.96 to $+1.96$, the p-value would exceed 0.05, suggesting that the comparative model could perform at the same level or better than the proposed LARSI-TPE-XGB. However, in this case, the results do not support this hypothesis. Similarly, we also replaced LARSI in our proposed LARSI-TPE-XGB with the Standard RSI for comparison. The resulting DM value of –2.135 remains statistically significant, indicating that the proposed LARSI provides a significant improvement in prediction. Therefore, our proposed LARSI-TPE-XGB is statistically superior to all of the benchmark models considered.

TABLE VIII
LARSI AND WEATHER EFFECTS ON LOAD FORECASTING METRICS FOR SWITZERLAND DATASET

Model	MAE	RMSE	MAPE
Basic-XGB	289.02	404.88	4.13%
LARSI-XGB	254.08	348.62	3.62%
LARSI-XGB with temperature	**249.06**	**337.03**	**3.57%**

Note: MAE and RMSE values are in **MW**.

As RF had a DM value of -4.14 and TPE-optimized RF (TPE-RF) had a similar DM value of -4.06, it is evident that TPE optimization did not have a significant effect. This can be attributed to RF requiring minimal hyperparameter tuning and its ability to perform well in high-dimensional spaces, even with default settings. However, XGBoost is more sensitive to hyperparameters, requiring careful adjustment of learning rate, regularization, sampling parameters, and other factors to achieve optimal performance. In high-dimensional spaces, the performance of XGBoost heavily depends on proper hyperparameter settings, which underscores the necessity of optimization techniques in certain scenarios.

F. Case IV: Impact of Weather Features on Load Forecasting

To further analyze the impact of LARSI and weather features on the load forecasting model, we utilized the Switzerland dataset. Due to the shorter time span of this data, a smaller validation set could result in poor generalization of the optimized hyperparameters. Therefore, we prioritized observing feature changes over applying TPE. In the data preprocessing phase, we incorporated temperature features from Switzerland's three major cities into the simulation; specifically, the maximum, minimum, and average temperatures (°F) for the next day.

Table VIII summarizes the results for the Switzerland dataset. The findings indicate that the proposed LARSI notably reduced errors, with an improvement of 12–13% across all metrics. However, incorporating temperature features resulted in only a slight error reduction. This may be attributed to the fact that Switzerland contains many cities, rendering temperature a less influential auxiliary feature. While short-term daily temperature fluctuations could be more useful for hourly forecasting, the focus on daily forecasting minimizes the impact of these variations. Over several days, temperature changes tend to be moderate, thus maintaining a marginal overall error reduction rate. As noted in [1], there is ongoing debate regarding the effectiveness of temperature as a forecasting feature. Some studies suggest that including temperature and precipitation data can enhance the accuracy of electricity demand forecasting, although concrete supporting evidence is not always presented. Our research offers new insights, suggesting that weather data may be more beneficial for predictions with finer granularity and more localized geographical coverage, such as in photovoltaic (PV) generation forecasting.

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V. Conclusion and Future Works

This study proposed a novel and efficient method, LARSI-TPE-XGB, for daily average load forecasting that effectively addresses key limitations of existing models. A significant challenge in load forecasting is the limited availability of long-term datasets, as power load data typically spans only a few years. This constraint makes it difficult to apply methods such as deep learning that rely on large datasets, particularly in the context of daily average load predictions.

To overcome this problem, we focused on developing effective feature extraction techniques and optimizing existing models. We proposed the Load Adaptive Relative Strength Index (LARSI), a concept derived from the Relative Strength Index (RSI) used in financial markets for short-term price predictions. By integrating LARSI with TPE and the XGBoost algorithm, our method enhanced both feature extraction and hyperparameter optimization. LARSI is specifically tailored for load forecasting, yielding an improvement of over 7% in prediction accuracy across three tested datasets. Additionally, TPE dynamically optimized the model's hyperparameters, complemented by time window update technology, allowing the model to adapt to evolving data without manual tuning. The LARSI-TPE-XGB resulted in a reduction of RMSE by 18.58% and 30.73% from two respective test datasets.

Comprehensive experimental results validated the generalizability and accuracy of LARSI-TPE-XGB. Analysis of feature importance and convergence were conducted, and the DM test results demonstrated its superiority over existing approaches. Our findings revealed that weather data has minimal impact on the accuracy of daily load forecasting, suggesting its greater utility in small-scale or high-resolution forecasts. In the future, we intend to extend LARSI-TPE-XGB for use in more complex scenarios to further improve load forecasting accuracy in other fields, such as incorporating renewable energy sources, thereby enhancing power system stability.

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